Examiner only

4. *(a)* Calcium and oxygen react forming calcium oxide.

(i) Explain, using dot and cross diagrams, how bonding takes place during the formation of calcium oxide. [2]

(ii)

Substance	Melting point (°C)
calcium oxide	2613
sodium chloride	801

Explain why both of these compounds have high melting points and why the melting point of calcium oxide is higher. [2]

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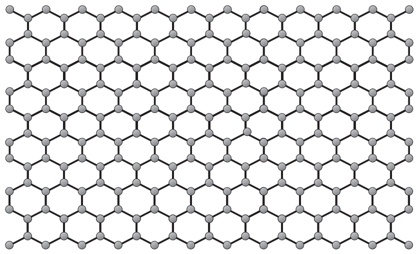
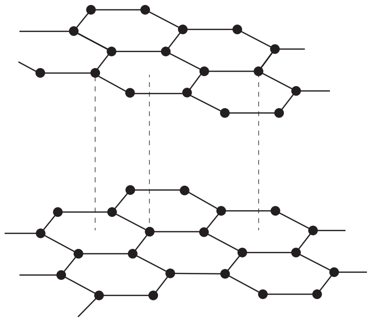
(b) Graphene and graphite are different forms of carbon.



graphene



graphite

(i) Explain how the bonding present in both forms makes them good conductors of electricity. [2]

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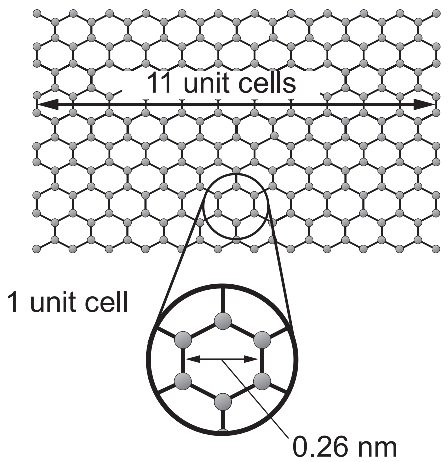
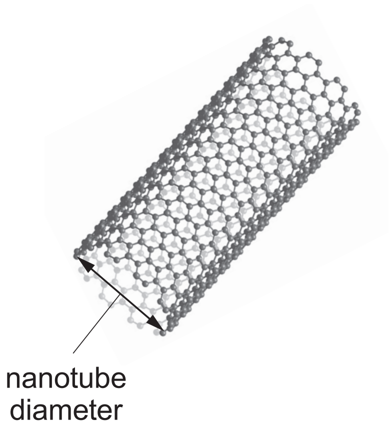
(ii) Carbon nanotubes can be formed by rolling up graphene sheets.





A sheet of graphene 11 unit cells in width was rolled up to form a nanotube. Use the information below to calculate the diameter, in metres, of the nanotube formed. Write your answer in **standard form**. [3]

$$\text{circumference} = \pi \times \text{diameter}$$

$$\pi = 3.14$$

$$1\ \text{nm} = 1 \times 10^{-9}\ \text{m}$$

Diameter = ... m

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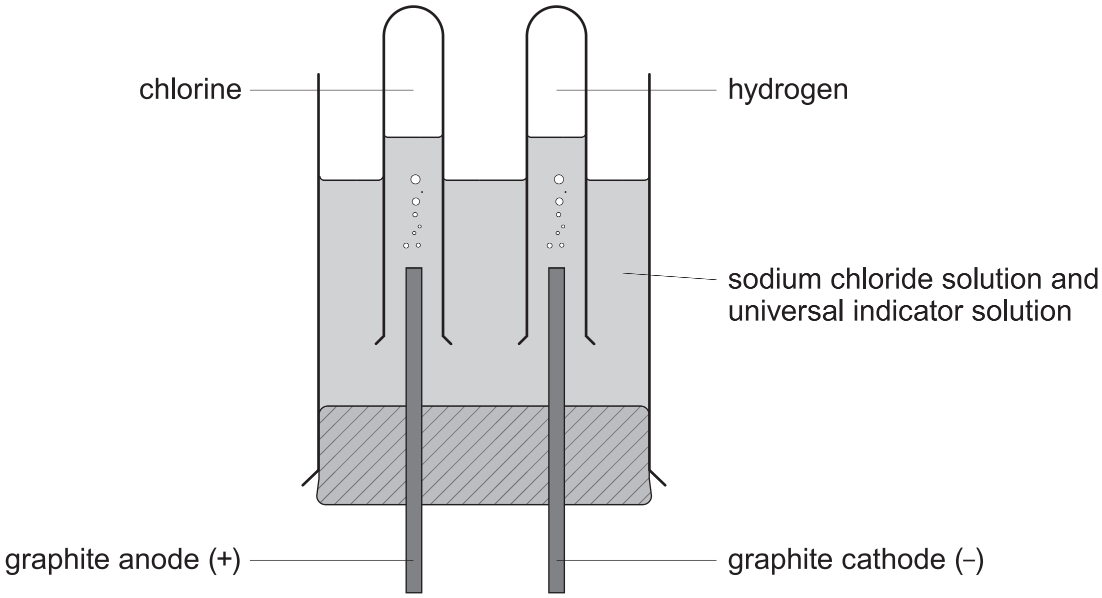
4. *(a)* The diagram shows apparatus that can be used for the electrolysis of sodium chloride solution.



(i) Explain why hydrogen, and **not** sodium, is formed at the cathode. [1]

....

....

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(ii) Complete the electrode equation for the reaction at the anode. [1]

$$2Cl^- \longrightarrow Cl_2 + \ldots\ldots\ldots\ldots$$

(iii) Explain why the universal indicator turns purple during electrolysis. [2]

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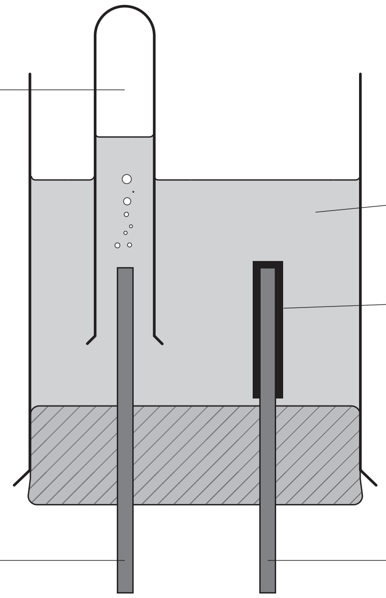
(b) The diagram shows the electrolysis of copper(II) sulfate solution.



(i) Explain, using the reaction occurring at the cathode, the meaning of the term *reduction*. [1]

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(ii) Over time the electrolyte turns from blue to colourless. State the change you would make to the apparatus so that the electrolyte remains blue during the process. Give a reason for your answer. [2]

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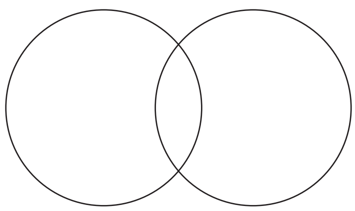
(iii) The electronic structure of oxygen is (2,6). Complete the diagram showing the outer shell electrons in an oxygen molecule, O_2. [2]



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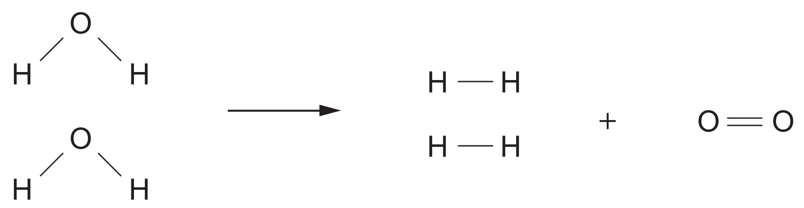
4. The equation shows the bonds which are broken and the bonds which are formed during the electrolysis of water.



(a) The total energy needed to break the bonds in the reactant is 1856 kJ.

Calculate the energy needed to break **one** O — H bond. [2]

Energy = ... kJ

(b) The total energy released when the bonds in the products are formed is 1370 kJ.

The bond energy for H — H is 436 kJ.

Calculate the energy released when making **one** O ═ O bond. [2]

Energy = ... kJ

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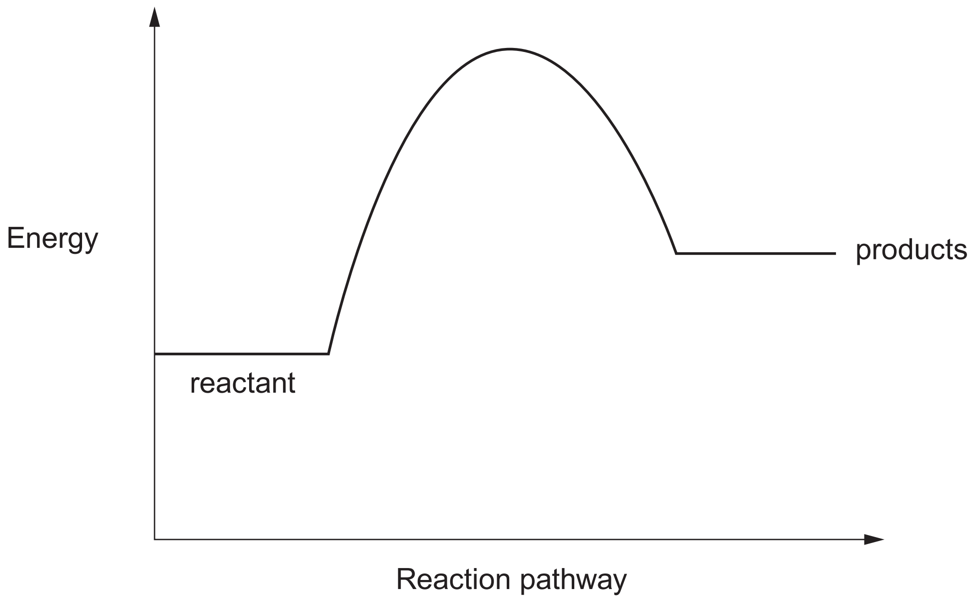
(c) The energy profile diagram shows the reaction to be endothermic. [1]

On the diagram below use the symbol ($\updownarrow$) to show the overall energy change for the reaction.



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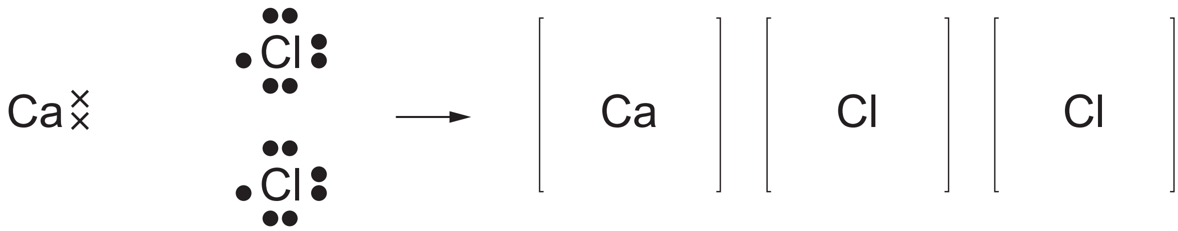
4. (a) Calcium reacts with chlorine to form the ionic compound calcium chloride.

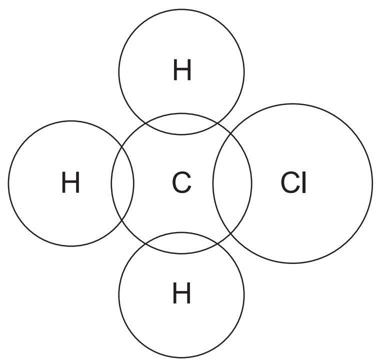
Complete the dot and cross diagram to show the electronic changes that take place when calcium reacts with chlorine to form calcium chloride.

Show the charge on each ion. [3]



(b) Complete the dot and cross diagram to show the bonding in a molecule of chloromethane. [2]

hydrogen 1 carbon 2,4 chlorine 2,8,7



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(c) Calcium and calcium chloride have different bonding and structures.

Explain how each substance conducts electricity.

You may include diagrams in your answer. [6 QER]

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4. (a) The table shows the results of an experiment in which zinc and lead powders were added separately to solutions of sodium chloride and iron(II) chloride.

	Sodium chloride solution	Iron(II) chloride solution
zinc	no change	colour change
lead	no change	no change

Use the results to place the four metals in order of reactivity. [1]

Most reactive ....

....

....

Least reactive ....

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(b) A student investigated metal reactivity using a different method.

He added 10.0 g of magnesium powder in 2.0 g portions to 50 cm^3 of zinc chloride solution and recorded the temperature of the mixture after each addition. He repeated the experiment with aluminium powder and again with copper powder.

The results for magnesium powder are shown below.

Mass of magnesium powder added (g)	Temperature (°C)
0	21.0
2.0	23.2
4.0	25.1
6.0	27.5
8.0	29.7
10.0	31.6

The results for aluminium and copper have been plotted on the grid opposite.

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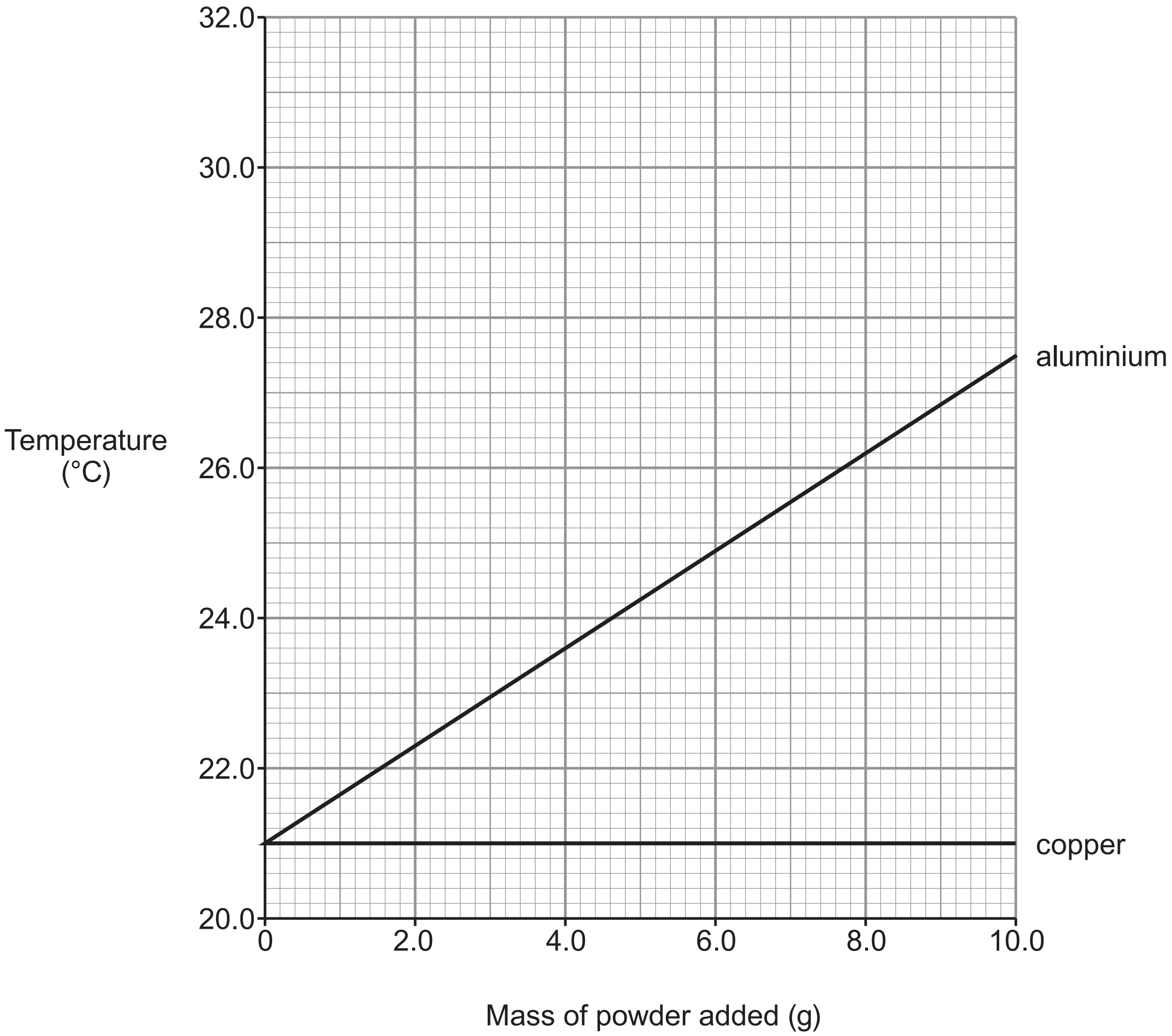
(i) Plot the results for magnesium on the grid. Draw a suitable line. [3]



(ii) State the conclusions that can be drawn about the reactivities of magnesium, aluminium, copper and zinc. Give your reasoning. [3]

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